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r/GooglePixel • 6mo ago
tarmachenry

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Seemingly Dangerous PWM - Google Pixel Videos and Tests

<https://ledstrain.org/d/3663-pixel-pwm-brutal>

Here is the link referenced at the source:

https://www.reddit.com/r/PWM_Sensitive/comments/177j74e/pixel_8_and_pixel_8_pro_pwm_videos_and_tests/

The Pixel 8 is a favorite smartphone device for a variety of reasons. However, the video at the link above shows it does not produce a stable, clean image, but instead uses a perverse evil called "pulse width modulation" to exhibit the screen at the desired brightness level. The awful flicker is not visually apparent but happens nevertheless. Surely this is not good for health?

How many of you have had symptoms from your use of Pixel devices?

I queried Google's AI:

"Yes, the severe Pulse Width Modulation (PWM) flickering on the Pixel 8a can cause eye strain, headaches, and other symptoms in people who are sensitive to it. The Pixel 8a has a low PWM frequency of 240Hz, which makes its OLED display particularly uncomfortable for flicker-sensitive individuals. Why PWM causes health issues

- **PWM brightness control:** Many OLED screens, including the Pixel 8a, use PWM to control brightness by rapidly switching the screen on and off. The frequency of this flicker determines its impact on sensitive users.
- **Physiological effects:** For some, even if the flicker is imperceptible, the brain must work harder to process the constantly changing light intensity. This can strain the eye muscles and lead to symptoms like:
 - Eye fatigue and dryness
 - Blurred vision
 - Headaches and migraines
 - Nausea and vertigo

Pixel 8a's specific issues

- **Very low frequency:** The Pixel 8a's 240Hz PWM is considered low and uncomfortable for sensitive people. For comparison, iPhones have historically used a 480Hz rate, and many newer phones use much higher frequencies (over 1000Hz) to reduce flicker.
- **Lack of alternatives:** Unlike some competing phones that offer a high-frequency flicker reduction mode, the Pixel 8a does not have a software setting to mitigate its low PWM rate.
- **Worse than predecessors:** The PWM rate on the Pixel 8a is even lower than its predecessor, the Pixel 7a, which had a rate of 360Hz."



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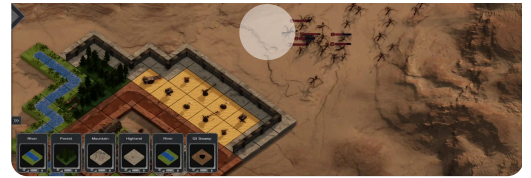
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Pixel 5 → 9 Pro

perverse evil called "pulse width modulation"

PWM isn't any kind of a "perverse evil", it's a perfectly valid technique. Some people are, supposedly, sensitive to direct PWM flicker, but this isn't some kind of evil tech.

[tarmachenny](#) **OP** • 6mo ago[\[deleted\]](#) • 6mo ago**IAmJakePaxton** • 6mo ago

Pixel 9 Pro

Is it normal for every launch cycle that this subreddit is flooded with wankers like this, who spew absolutely nonsensical negative stuff about Pixel and Google?

This is after the other guy who said "hundreds" of pixels are at risk of bricking (if users go and install custom roms)

**Grunslk** • 6mo ago

Pixel 8a

This link refers to the comments on a post on another subreddit. What kind of tinfoil hat garbage are you on about?

[tarmachenny](#) **OP** • 6mo ago • Edited 6mo ago**tarmachenny** **OP** • 6mo ago

Please avoid attacking the messenger and instead focus your efforts on the content at hand.

"Does anyone else have these symptoms? OLED screens ruin my head 100x worse, for example. I can't stand them, even from far away..."

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Pixel 8a

We're attacking anecdotal fear-mongering. Presenting a supposed quote that isn't even attributed to a named person along with a YT video and the comment section of a reddit post is not "content," except in the comedic sense.

**tarmachenny** OP • 6mo ago • Edited 6mo agoPixel 8a: <https://youtube.com/watch?v=aoNSYNPbCtE>

"This is an extremely uncomfortable display" (summary of the measurements at the end of this video)

Are you willing to take a picture of your eyes after using your Pixel 8a so we can see for ourselves the redness and soreness? I've read of some having bags under their eyes that go away when they switch from OLED to LCD. Consider too if it is doing that to your eyes, what is it doing to your brain?

Also added an AI commentary to my original post.

**Grunslík** • 6mo ago

Pixel 8a

More YT videos? Really?

**tarmachenny** OP • 6mo ago

From AI:

"Unlike the Moto Edge 2025 and Pixel 9a, the Moto G 2025 uses an LCD screen with DC dimming, which is a much gentler method of brightness control than the PWM dimming used by most OLED screens."

"For some users, particularly those sensitive to Pulse Width Modulation (PWM) flicker, the eye strain from the Pixel 8a is described as "horrible" or "extremely uncomfortable". On [Reddit](#) and other forums, multiple users have detailed their struggle with the phone, reporting symptoms like eye strain, headaches, and nausea."

Why the eye strain can be so severe

- **Low PWM frequency:** The Pixel 8a has a very low PWM frequency of 240Hz, which is used to dim its OLED screen. For comparison, some competing phones now use frequencies over 1000Hz, and even iPhones have a higher rate. A lower frequency is more likely to be perceived as a flicker by sensitive individuals.
- **Perceived flickering:** Users sensitive to PWM often describe the screen as "flickering," "pulsating," or causing their vision to feel "weird". This is because the eye and brain, even if not consciously aware of the flicker, have to work harder to compensate for the rapid changes in brightness.
- **Strong reactions:** Some users have reported experiencing symptoms almost immediately upon using the phone, describing their eyes as "killing" them and getting "excruciating" headaches. These



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**DoctorSora** • 5mo ago

I used the Pixel 9 Pro XL for a few days, and it hurt my eyes like crazy. I sold it immediately. Nowadays, I am using the Pixel 8 Pro, and my eyes feel normal.

**Mat25_AT** • 1mo ago

Funny, I just came from the Pixel 6 Pro with no problems at all to the Pixel 8 Pro and can't use it for longer than 5 minutes becoming problems like tinnitus and feeling uncomfortable.